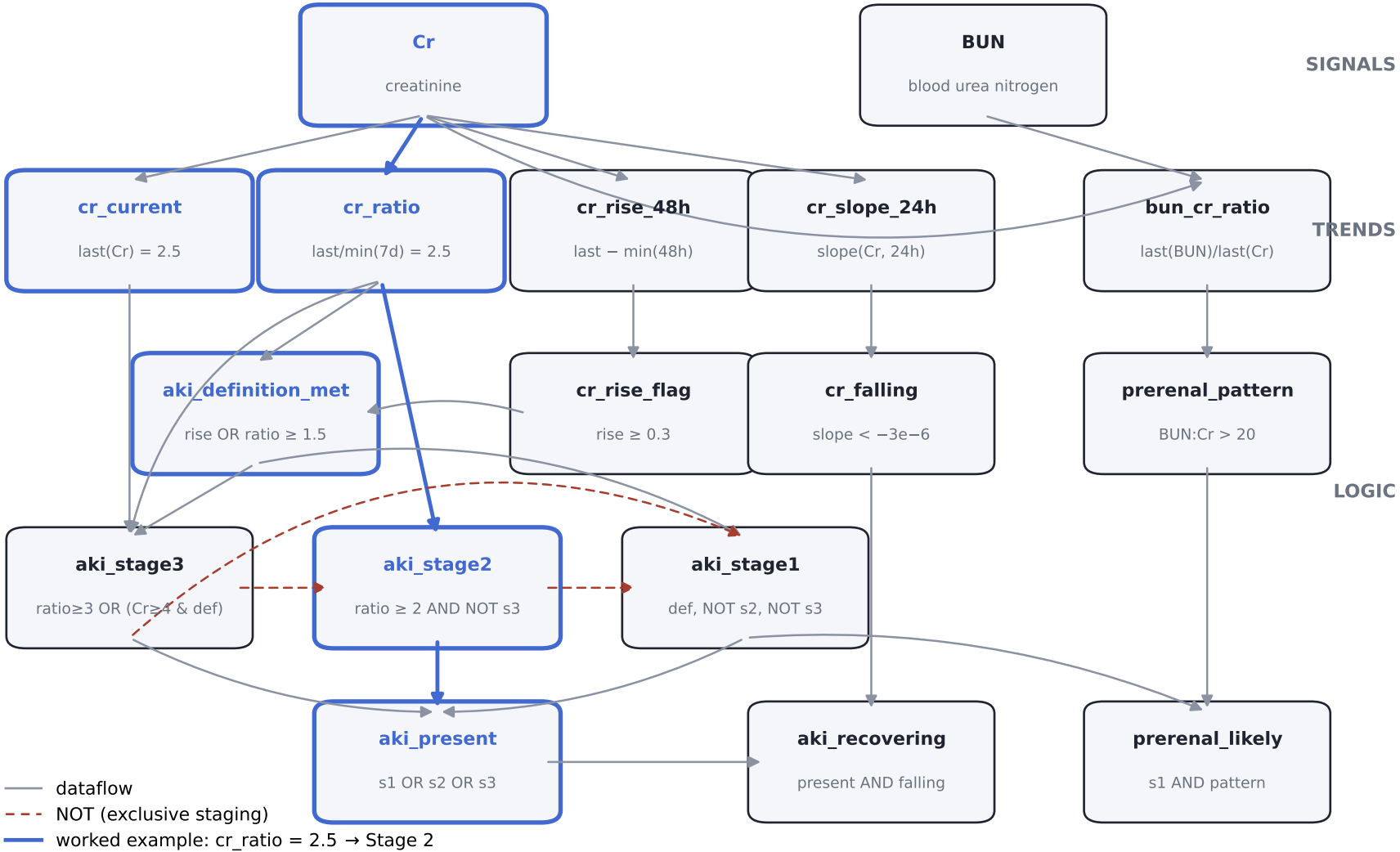
Cr
creatinine
BUN
blood urea nitrogen
SIGNALS
cr_current
last(Cr) = 2.5
cr_ratio
last/min(7d) = 2.5
cr_rise_48h
last − min(48h)
cr_slope_24h
slope(Cr, 24h)
bun_cr_ratio
last(BUN)/last(Cr)
TRENDS
aki_definition_met
rise OR ratio ≥ 1.5
cr_rise_flag
rise ≥ 0.3
cr_falling
slope < −3e−6
prerenal_pattern
BUN:Cr > 20
LOGIC
aki_stage3
ratio≥3 OR (Cr≥4 & def)
aki_stage2
ratio ≥ 2 AND NOT s3
aki_stage1
def, NOT s2, NOT s3
aki_present
s1 OR s2 OR s3
aki_recovering
present AND falling
prerenal_likely
s1 AND pattern
dataflow
NOT (exclusive staging)
worked example: cr_ratio = 2.5 → Stage 2